Karl Kunze, Ph.D.

Plant Breeder — Genetics, Genomics & Quantitative Breeding

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EDUCATION

Cornell University | Ithaca, NY — Doctorate degree Completion: 06/2023

Major: Plant Breeding and Genetics | Minor: Plant Pathology and Food Science

PLBRG 7160 Perspectives in Plant Breeding (3 cr) · PLBRG 7170 Quantitative Genetics in PLBRG (3 cr) · BRTY 6830 Quantitative Genomics and Genetics (4 cr) · BIOMG 4810 Population Genetics (3 cr) · BIOMG 4000 Genomics (3 cr)

Cornell University | Ithaca, NY — Bachelor's degree Completion: 05/2017

Major: Plant Science | Minor: Plant Breeding and Genetics

BIOMG 2800 Lectures in Genetics and Genomics (3 cr) · PLBRG 4060 Methods of Plant Breeding Lab (2 cr) · PLBRG 4070 Crop Nutritional Quality (2 cr) · PLBRG Genetic Improvement of Crop Plants (3 cr) · BIOPL Molecular Biology and Genetic Engineering (3 cr) · PLBRG 2250 Plant Genetics (4 cr)

Student Exchange Program, Wageningen University Research, Wageningen, NL — January-June 2016

LANGUAGE SKILLS

Spanish Spoken: Novice Written: Novice Read: Intermediate

PROFESSIONAL EXPERIENCE

Plant Breeder I, Hot Pepper | *HM Clause/Limagrain Vegetable Seeds* | *Davis, CA* 03/2024 - Present *(45 hrs/wk)*

- Responsible for delivery of new competitive hybrids from our fresh market jalapeño breeding program, target market value approximately $40,000,000 USD
- Evaluated over 450 jalapeño hybrids at 4 internal site-location trials per year (2 times each at Davis, CA and Los Mochis MX.; 3 times at Felda, FL in Spring and once in Immokalee, FL in Fall)
- Routinely evaluated 200 hybrids at 3 locations of jalapeño, serrano, ancho, banana and anaheim in Bajio, MX once a year (3 times since start date). Data collected includes visual data collection, harvested fruit yield and implementation of internal image detection pipeline.
- Developed approximately 45 new crosses per year across two greenhouse cycles for population development of improved fresh Jalapeno lines
- Advance inbred lines of jalapeño using combination of internal DH pepper line development and pedigree based approaches
- Used genetic data of 400+ inbred lines and phenotypic data of hybrid trials of jalapeño to develop general combining ability values to select new inbred parent lines for crossing, as well as develop prediction models for new hybrids
- Used a combination of Affrymetrix genotype data and phenotype data across all hot pepper typologies (1,100+ lines) to develop structured heterotic breeding pool structure for our jalapeño, serrano, banana, and anaheim germplasm
- Developed systematic naming conventions and routine database updating procedures to increase accuracy of relevant data in internal breeding database systems
- Collected data and co-authored a comprehensive "Plant breeding review" of the hot pepper breeding program in HM Clause, presented to internal management and colleagues, with plans for new strategies to increase program efficiency
- Collaborated with internal stakeholders of product development specialists and research trial specialists in Mexico to evaluate and advance hybrid varieties for potential commercialization. Includes developing procedures and trial summaries that encourage use of evidence based decision making on quantitative data to determine hybrid advancement

- Attended an internal Limagrain research conference in Barcelona, Spain in January 2025, presented a poster discussing use of machine harvest in jalapeños
- Attended an internal Limagrain pepper crop committee meeting in Almeria, Spain in November 2025, presented on progress and roadblocks of internal image analysis pipeline of jalapeños

Post Doctoral Research Associate | *Cornell University* | *Ithaca, NY* 07/2023 - 02/2024 *(40 hrs/wk)*

- Collaborated on a Foundation for Food and Agriculture research grant (AFRI) with colleagues at Cornell University, New Mexico State University and Virginia Polytechnic Institute to analyze systematic aerial imaging and ground truth data of repeated harvest alfalfa trials
- Performed logistical and data management of multi-spectral imaging of a total of 56 flights conducted in Ithaca, NY and 26 flights conducted in Las Cruces, NM. The trial in Ithaca, NY consisted of 36 alfalfa entries harvested 3 times in a season across two years. The trial in Las Cruces, NM consisted of 24 alfalfa varieties harvested 8 times in one year.
- Used novel statistical methods such as random regression and spatial analysis to model alfalfa growth curves using extracted NDVI values as a proxy for ground truth biomass
- Experienced using open-source database breeding software such as Imagebreed and Breedbase developed at the Boyce Thompson Institute, and interfaced with software developers to improve usability for daily users

Graduate Research Assistant/PhD Candidate | *Cornell University* | *Ithaca, NY* 08/2017 - 06/2023 *(55 hrs/wk)*

- Advanced winter malting barley double haploid and recombinant inbred populations starting with crosses in 2017 resulting in the release of variety "Lake Effect" in September 2025
- Using the DH line population, conducted an extensive seed dormancy assay experiment over two years to determine dormancy loss and characterize genetic marker trait associations associated with barley germination. A total of 450 lines were measured at 3 locations over two years, and an average of 5 timepoints per year were conducted.
- Micro malted 240 winter barley over two time points at the USDA Cereal Crops and Research Unit in Madison, WI. Malting quality results were used to inform selection decisions.
- Used genomic prediction, linear mixed models, spatial correction and genome wide association studies to advance lines and inform selection decisions for barley line selection and advancement
- Collaborated with Oregon State University, UW-Madison and University of Minnesota under the organic multi-use naked barley project funded by the USDA Organic Agricultural Research and Extension Initiative competitive grants program for 6 years after initial project approval and renewal
- Participated in outreach events such as demonstration of barley to K-12 programs, participation of "variety showcase events" connecting organic farmers/plant breeders to bakers, chefs, etc., as well as developing webinars and short publications on eOrganic
- Performed genotype by environment analysis for eight organically managed winter naked barley trials across the northern United States
- Performed a genome wide association analysis of 250 diverse barley lines across 13 environments for foliar disease resistance in natural field organic conditions
- Used aerial imaging of a spring naked barley variety trial over two years to quantify barley growth over time as a precursor for competition with weeds

PUBLICATIONS

1. Kunze, K. H., Rooney, T. E., Sweeney, D., & Sorrells, M. E. (2026). The SD1 locus affects primary seed dormancy in winter malting barley. *The Plant Genome*, 19, e70234. doi.org/10.1002/tpg2.70234
2. Kunze, K. H., Meints, B., Massman, C., Gutiérrez, L., Hayes, P. M., Smith, K. P., Bergstrom, G. C., & Sorrells, M. E. (2024). Genome-wide association of an organic naked barley diversity panel identified quantitative trait loci for disease resistance. *The Plant Genome*, 17, e20530. doi.org/10.1002/tpg2.20530
3. Kunze, K. H., Meints, B., Massman, C., Gutiérrez, L., Hayes, P. M., Smith, K. P., & Sorrells, M. E. (2024). Genotype × environment interactions of organic winter naked barley for agronomic, disease, and grain quality traits. *Crop Science*, 64, 678–696. doi.org/10.1002/csc2.21195

4. Thapa, R., **Kunze, K. H.**, Hansen, J., Pierce, C., Moore, V., Ray, I., ... & Robbins, K. (2024). Remote sensing for estimating genetic parameters of biomass accumulation and modeling stability of growth curves in alfalfa. *G3: Genes, Genomes, Genetics*, 14(11), jkae200. doi.org/10.1093/g3journal/jkae200
5. Massman, C., Meints, B., Hernandez, J., **Kunze, K.**, Smith, K. P., Sorrells, M. E., Hayes, P. M., & Gutierrez, L. (2023). Genomic prediction of threshability in naked barley. *Crop Science*, 63, 674–689. doi.org/10.1002/csc2.20907
6. Massman, C., Meints, B., Hernandez, J., **Kunze, K.**, Hayes, P. M., Sorrells, M. E., Smith, K. P., Dawson, J. C., & Gutierrez, L. (2022). Genetic characterization of agronomic traits and grain threshability for organic naked barley in the northern United States. *Crop Science*, 62, 690–703. doi.org/10.1002/csc2.20686
7. Rooney, T. E., **Kunze, K. H.**, & Sorrells, M. E. (2022). Genome-wide marker effect heterogeneity is associated with a large effect dormancy locus in winter malting barley. *The Plant Genome*, 15, e20247. doi.org/10.1002/tpg2.20247
8. Rooney, T.E., Sweeney, D.W., **Kunze, K.H.** et al. Malting quality and preharvest sprouting traits are genetically correlated in spring malting barley. *Theor Appl Genet* 136, 59 (2023). doi.org/10.1007/s00122-023-04257-6
9. Bunting, J. S., Ross, A. S., Meints, B. M., Hayes, P. M., **Kunze, K.**, & Sorrells, M. E. (2022). Effect of Genotype and Environment on Food-Related Traits of Organic Winter Naked Barleys. *Foods*, 11(17), 2642. doi.org/10.3390/foods11172642
10. Sweeney, D.W., **Kunze, K.H.** & Sorrells, M.E. QTL x environment modeling of malting barley preharvest sprouting. *Theor Appl Genet* (2021). doi.org/10.1007/s00122-021-03961-5

In Review

- Sepp, Siim Samuel; Kunze, Karl; Bergstrom, Gary; Kolkman, Judy; Hayes, Patrick; Sorrells, Mark E. "Registration of Winter malting barley LakeEffect". *Journal of Plant Registrations*. Submitted revisions June 2026.
- Kolkman, J.M., Sepp, S.S., **Kunze, K.**, Bergstrom, G.C., & Sorrells, M.E. Genome wide association analysis of resistance to scald in an adapted multiparent winter malting barley population. *bioRxiv* 2026.03.12.711358; doi.org/10.64898/2026.03.12.711358

Non-Academic Publications

- Bradley, E. A., & Kunze, K. (2023). Building Skills: Utilizing Software for Data Analysis and Professional Presentations. *CSA News*, 68(7), 36–38.

CONFERENCE PRESENTATIONS

- Moderated a special session at CANVAS titled "Transitioning from Graduate School to Early Career" — November 2023, St. Louis, MO
- Served as moderator of Crop Breeding and Genetics Oral III and presented a conference talk at CANVAS on "Genotype by environment interaction of organic winter organic naked barley" — November 2022, Baltimore, MD
- Presented a conference talk at CANVAS on "Interaction of Pre-Harvest Sprouting, Germination Rate and Malting Quality for Winter and Facultative Malting Barley" — November 2022, Baltimore, MD
- Presented a conference talk at CANVAS on "Utilizing Aerial Imaging for Components of Weed Competitive Ability in Spring Naked Barley" — November 2021, Salt Lake City, UT

PROFESSIONAL SERVICE & COMMITTEE INVOLVEMENT

- Member, Crop Science Society of America, 2018–2020, 2021–2024
- Graduate Student Board Representative, Board of Directors of the Crop Science Society of America, January 2021–December 2023; participated in Science Policy sub-committee
- Member, American Society of Agronomy (ASA) / Crop Science Society of America (CSSA) / Soil Science Society of America (SSSA) Graduate Student Working Group, January 2021–December 2023
- President, Cornell Plant Breeding and Genetics graduate student organization, Synapsis, August 2018–August 2019